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Enzymatic and acidic degradation of high molecular weight dextran into low molecular weight and its characterizations using novel Diffusion-ordered NMR spectroscopy

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ABSTRACT

Low molecular weight fractions were derived from native high molecular weight dextran produced by *Leuconostoc mesenteroides* KIBGE-IB26. Structural characterization of native and low molecular weight fractions obtained after acidic and enzymatic hydrolysis was done using FTIR and NMR spectroscopy. The molecular weight was estimated using Diffusion Ordered NMR spectroscopy. Native dextran (892 kDa) is composed of α-(1 → 6) glycosidic linkage along with α-(1 → 3) branching. Major proportion of 528 kDa dextran was obtained after prolong enzymatic hydrolysis however, an effective acidic treatment at pH-1.4 up to 02 and 04 h of exposure resulted in the formation of 77 kDa and 57 kDa, respectively. The increment in pH from 1.4 to 1.8 lowered the hydrolysis efficiency and resulted in the formation of 270 kDa dextran fraction. The results suggest that derived low molecular weight water soluble fractions can be utilized as a drug delivery carrier along with multiple application relating pharmaceutical industries.

1. Introduction

Several native high molecular weight exopolysaccahrides (EPSs) beside their multiple industrial applications are hydrolyzed into various low molecular grade fractions in order to explore these fractions for more concise and accurate consumption in different industries. Bacterial EPSs are preferably used because of their unique structural diversity and texture. These EPSs can be used as a carrier molecule in pharmaceutical industries for drug delivery beside viral carrier molecules. However, studies have shown that non-viral vectors are not efficient enough but still provide safer drug delivery as compared to viral vectors. Natural polysaccharides which are used as an effective drug delivery carrier must be biodegradable in nature. For this purpose, most widely recognized polymers among pharmaceutics include pectin, cyclodextrin, amylose, chitosan, guar gum, chondroitin sulphate and inulin [1–3]. Temperature sensitive hydrogels are also a unique selection for this purpose [4]. The utilization of natural EPSs depends upon its molecular weight and the nature of the chemical groups present in it. Dextran is a neutral polysaccharide, consists of D-glucopyranoside residues linked in a linear manner with α-(1 → 6) glycosidic linkages and may also have α-(1 → 2), α-(1 → 3) and α-(1 → 4) branch points. Beside several industrial applications of high molecular weight dextran, some of its low molecular weight fractions have gained wide attention for pharmaceutical purposes. Low molecular weight dextran having molecular mass ranging from 5.0 kDa to 100 kDa, is used for various clinical purposes. This water soluble EPS, that has a biodegradable nature, is used as clinical grade dextran, which is commercially available as Dextran-1, Dextran-40, Dextran-60 and Dextran-70. These fractions are used as intravenous injectable to replace blood loss, substitute blood plasma levels and for the management of anticoagulant therapy [5]. They are also exploited as cryopreservatives for transplant organs and as carrier molecule in vaccines [6].

Native dextran is produced as high molecular weight polymer by multiple microbial sources, which are not suitable to be used in medical context because of their high level of toxicity and

Abbreviations: EPS, exopolysaccharide; FTIR, Fourier-transform infrared; SEM, scanning electron microscopy; NMR, nuclear magnetic resonance spectroscopy; 2D NMR, two dimensional nuclear magnetic resonance spectroscopy; DOSY, diffusion-ordered spectroscopy; COSY, homonuclear correlation spectroscopy; TOCSY, total correlation spectroscopy; HSQC, heteronuclear single-quantum correlation spectroscopy; HMBC, heteronuclear multiple-bond correlation spectroscopy.

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immunogenicity. After their hydrolysis by either chemical or enzymatic treatment, the low molecular weight fractions were tested and analyzed for their cytotoxic effect in the last few decades. Most recent studies have focused towards the development of an effective drug delivery polymer as a carrier vector molecule. The structure-function-cytotoxicity relationship of low molecular weight dextran fractions as dextran-peptide hybrid system was studied recently [7]. They suggested Dex10-R5H5 and Dex20-R5H5 hybrid induce high gene expression level with low cytotoxicity as compared to Dex70-R5H5. Commercially available dextran (T10, T50, T70, T500 and T2000) was also lately analyzed for the antiviral property against salmonid viruses [8]. However, this study revealed that T2000 was found to be more effective as compared to lower molecular weight fractions.

The present study is an effort to produce multiple low molecular weight fractions using chemical and enzymatic hydrolysis of a native high molecular weight dextran produced by *Leuconostoc mesenteroides* KIBGE-IB26. For the structural characterization of the hydrolyzed dextran different analytical techniques were employed.

2. Experimental

2.1. Chemical used

All chemicals used in this study were of analytical grade. Dextranase was obtained from *Bacillus megaterium* KIBGE-IB31 [GenBank: KF241867].

2.2. Cultivation and taxonomic characterization of microorganism used

Bacterium used in this study was isolated from rotten vegetable sample of *Lagenaria siceraria* (bottle gourd). Isolation and identification of *Leuconostoc mesenteroides* KIBGE-IB26 [GenBank: KF241862] was carried out according to the method described earlier [9]. Identification was further confirmed by 16S rDNA sequence analysis and the culture was maintained on tomato juice agar slant (pH 7.2) at 4 °C [9].

2.3. Production and purification of dextran

Native dextran was produced by inoculating the culture in a sterile medium containing (g L^{-1}): sucrose, 100.0; peptone, 5.0; yeast extract, 2.5; K_2HPO_4, 20.0; $CaCl_2{\cdot}2H_2O$, 0.025; $MgSO_4{\cdot}7H_2O$, 0.01; $MnCl_2{\cdot}H_2O$, 0.01 and NaCl, 0.01 with a pH adjusted to 7.5 before sterilization. For dextran production, 10^6 cells ml^{-1} were transferred in production medium and incubated at 25 °C for 18.0 h under static condition. After fermentation, the dextran was precipitated and purified according to the previously described method [10]. The results presented are the mean values of triplicate experimentations n = 3, S.E. 5.0%, p-value: < 0.005.

2.4. Optimization of fermentation parameter for native dextran production

Physical and chemical fermentation parameter that affect the process were optimized including incubation temperature, medium pH, substrate concentration and fermentation time for *L. mesenteroides* KIBGE-IB26 by means of one parameter at a time variation strategy (Table 1).

2.5. Hydrolysis of high molecular weight dextran

For the production of low molecular weight dextran fractions, native dextran was hydrolyzed by enzymatic and acid treatments:

Table 1
Optimization physical parameters of fermentation for native dextran production.

Fermentation variables	Dextran (g)	% Conversion [a]
Time course [b] (h)		
6	4.65 ± 0.23	46.5
18	4.93 ± 0.24	49.3
24	4.21 ± 0.21	42.1
36	4.0 ± 0.20	40.0
48	2.8 ± 0.14	28.0
Temperature [c] (°C)		
15	1.54 ± 0.07	15.4
20	2.34 ± 0.11	23.4
25	4.93 ± 0.24	49.3
30	1.03 ± 0.05	10.3
35	0.0	0.0
pH [d]		
6.5	4.03 ± 0.20	40.3
7.0	4.21 ± 0.21	42.1
7.5	4.93 ± 0.24	49.3
8.0	4.62 ± 0.23	46.2
8.5	4.15 ± 0.20	41.5
Sucrose [e] (g L^{-1})		
50.0	20.9 ± 1.04	41.8
100.0	49.3 ± 2.40	49.3
150.0	64.6 ± 3.20	43.1
200.0	58.8 ± 2.94	29.4
250.0	59.2 ± 2.96	23.7
300.0	54.2 ± 2.71	18.0

[a] Calculation based on conversion of initial concentration of sucrose into dextran.
[b] Dextran produced using 10.0 g dL^{-1} sucrose medium with initial pH 7.5 at 25 °C.
[c] Dextran produced using 10.0 g dL^{-1} sucrose medium with initial pH 7.5.
[d] Dextran produced using 10.0 g dL^{-1} sucrose medium and pH of the medium adjusted before autoclaving.
[e] Dextran produced at 25 °C with medium pH 7.5.

2.5.1. Acid hydrolysis

Native dextran solutions were prepared (20.0 g L^{-1}) in deionized water and the pH value was adjusted to 1.4 and 1.8 by the addition of sulphuric acid (0.1 N) and kept at 80 °C for different time intervals [11]. Dextran solutions were precipitated by chilled ethanol with continuous stirring (200 rpm) and kept at 4 °C for equilibration for 24.0 h. Precipitated dextran was separated by centrifugation at 40248 × g for 10 min. All the samples were completely dried and stored at 30 °C for further analysis.

2.5.2. Enzymatic hydrolysis

For enzymatic hydrolysis, dextran sample (10.0 g L^{-1}) was prepared in citrate phosphate buffer (0.1 M, pH-4.5) and dextranase (250U) was added. Reaction mixture was kept at 50 °C up to 72 h in shaking incubator (75 rpm). After 06, 12, 24, 48 and 72 h intervals dextran was re-precipitated by the aforementioned method. Samples were stored at 30 °C for further examination.

2.6. Purification of hydrolyzed product

Purification for both acidic and enzymatic hydrolyzed fractions were performed by passing the dextran solutions through different molecular weight cutoff filter membranes (Ultracel, PLC membrane 30 kDa,100 kDa,300 kDa) to obtain various clinical grade fractions. The fractionated samples were further characterized by Fourier-transform infrared (FTIR), ^{1}H NMR, ^{13}C NMR and 2D NMR (HSQC and HMBC) spectroscopy.

2.7. Analytical procedures

Total protein content [12], total carbohydrate content [13] and reducing sugar [14,15] were quantified using methods described earlier. Optical rotation of the purified dextran solution was estimated using ATAGO Polax 2L at 25 °C. The results presented are the mean values of triplicate experimentations n = 3, S.E. 5.0%, p-value: < 0.001.

2.8. Scanning electron microscopy (SEM)

Dried samples of native and hydrolyzed dextran fractions were analyzed for surface topography by using SEM. All the experimental conditions and protocols were followed as described earlier [16].

2.9. End product analysis of dextran

Thin layer chromatography was used to determine the formation of end products after treatment with dextranase [17,18]. Dextranase (250 Units) was obtained from *Bacillus megaterium* KIBGE-IB31 [GenBank: KF241867] and was amalgamated with dextran solution (10 g L^{-1}; 0.1 M citrate phosphate buffer; pH-4.5). The reaction mixture was kept at 50 °C up to 120 h and samples were collected after different time intervals and kept in boiling water bath for 10 min. Standards (Sigma: glucose, isomaltose, isomaltotriose, isomaltopentose, isomaltoheptose) and samples were analyzed on silica coated plates (Silica gel 60, Merck, Germany) using *n*-butanol: acetic acid: water (5:3:1). The results were monitored by spraying 20.0% sulfuric acid and spots were visualized after placing the plates at 110 °C.

2.10. Determination of molecular weight and structure of dextran

2.10.1. NMR spectroscopy

Average molecular weight of dextran samples were determined by using Diffusion-ordered NMR spectroscopy (DOSY). DOSY experiments were carried out at 300 K on a Bruker 600 DRX equipped with a cryo-probe. All the experiments were carried out according to the method described earlier by slight modification [16]. Typically, 800 mg of each native and hydrolyzed fraction were dissolved in 700 ml of D_2O. The stebpgp1 s pulse sequence from Bruker library, incorporating bipolar gradient pulse pair and 1 spoil gradient, with a longitudinal eddy current delay, was run with linear gradient (5.35 G cm^1) stepped between 2.0% and 95.0%, incremented in 32 steps. The gradient duration (δ) was held constant at 2 ms and the echo delay (Δ) at 100 ms. After Fourier transformation and baseline correction of 1D ^{1}H spectra (F2 dimension), the diffusion dimension (F1) of the 2-dimensional DOSY was processed by using TopSpin software (version 2.1).The calibration curve, derived from the study of pullulan fractions of known molecular weights, reported by Viel and coworkers (2003), is as follow:

$$\log D = -0.484 \log MW - 8.114.$$

2.10.2. Fourier transform- infrared spectrometric (FTIR) analysis

FTIR spectrum of native dextran and purified low molecular weight dextran samples (obtained after hydrolysis) were recorded as described previously [9,16].

3. Results and discussion

Dextran is a unique exopolysaccharide which has high shear rate and water solubility. High molecular weight dextran is used in transglycosylation reactions of low molecular weight drugs [19]. Due to toxicity levels, high molecular weight dextran cannot be used in clinical context until it is hydrolyzed into low molecular weight fractions [20–22]. Native dextran can hydrolyzed using dilute mineral acids however; the data for dextran hydrolysis is still scarce in this field. Limited reports are available on hydrolysis of native dextran to produce clinical grade fractions [11,23–26]. Several fractionation procedures have been suggested previously using modern techniques including immobilization of dextranase but due to cost effectiveness industries prefer acid hydrolysis [27]. Keeping these points in view, the current study is designed to produce low molecular weight dextran fractions using mineral acid and dextranase treatment.

Native dextran was produced using *L. mesenteroides* KIBGE-IB26 in batch fermentation in a designed medium containing sucrose as a substrate (Table 1). Initially, all the fermentation parameters were optimized. In the first step of optimizing the operating conditions for dextran production, fermentation time was monitored up to 48 h. Highest yield of native dextran was observed at 18 h with 49.3% conversion of sucrose into dextran polymer. *L. mesenteroides* AA1, *L. mesenteroides* KIBGE-IB22M20 and *L. mesenteroides* KIBGE-IB22 were reported to produce native dextran at 12, 18 and 24 h of fermentation with a percent conversion of 48.9, 43.7 and 24.1, respectively. *L. mesenteroides* NRRL B-512(F) produced native dextran of approximately 39 million daltons after 23 h of incubation but when the fermentation time was prolonged up to 273 h a remarkable decrease in the approximate molecular mass was observed without any change in the percent yield of the polymer [28,29]. Hence, fermentation time is also an important criterion reported earlier for the production of specific polymer size. *L. mesenteroides* KIBGE-IB26 is able to cultivate at 25 °C, with a pH of 7.5, with maximum percent conversion of 100 g L^{-1} sucrose into dextran. Fermentation parameters affect the recovery as well as the molecular weight of the native dextran produced. Various researchers have reported different mechanisms that affect the size and the yield of synthesized native dextran with reference to fermentation temperature. Some of them suggested that temperature could increase the polymerization kinetics of the transglycosylation reaction of dextranases [30]. The synthesized dextran become an acceptor polymer and thus helps in the binding of other short chains with the main backbone α-(1 → 6) of the dextran resulting in the increase of the polymer size by increasing the α-(1 → 3) branch linkage [31]. Another most critical parameter is the substrate concentration. As the concentration of sucrose is elevated, an increase in the dextran yield is observed suggesting that the concentration of substrate is directly proportional with the production of the polymer [30]. However, there is another concept linked with the substrate, which at higher concentrations may act as an inhibitor for the polymerization of glucose molecule in to native dextran chain [9].

The molecular weight of the native dextran produced by *L. mesenteroides* KIBGE-IB26 was found to be 892 kDa (D1) (Table 2). Initially, for the hydrolysis of this native polymer, dextranase was employed and the molecular mass of the fractions were determined using diffusion ordered spectroscopic (DOSY) analysis (Fig. 1A and 1B). This enzymatic hydrolysis resulted in the fractionation of 528 kDa (D7) dextran when exposed to the enzyme up to 72 h. This molecular weight fraction obtained was not in the range of clinical grade dextran therefore, any further hydrolysis by enzymatic means was not favorable as the dextranase lost is catalytic potential during this incubation time period. Dextranase from other species have also been used to hydrolyze dextran [32]. The dextranase used in the current study is produced from *B. megaterium* KIBGE-IB31, which is stable and cost effective as compared to commercially available dextranases [33].

In the current study, for further fractionation of the synthesized native dextran into low molecular fractions, the native dextran was exposed to dilute mineral acid (0.1N H_2SO_4); though, this type of hydrolysis causes non-specific cleavage and results in the formation of polydispersed products. The low molecular weight dextran that were produced after the treatment with sulphuric acid, with a pH value of 1.4, for 2 and 4 h were approximately 77.0 kDa and 57.0 kDa, respectively. However, when the pH was raised from 1.4 to 1.8, the acidic treatment was not as effective as lower pH

Table 2
Chemical and enzymatic hydrolysis of native dextran into different low molecular weight fractions produced by *Leuconostoc mesenteroides* KIBGE-IB26.

Sample	pH[a]	Temperature[b] (°C)	Time[c] (hours)	Molecular weight (kDa)	Reference
D1[d]	nd	nd	nd	892.0	Current study
Acid Hydrolysis					
D4[e]	1.4	80.0	2.0	77.0	Current study
D5[e]	1.4	80.0	4.0	57.0	Current study
D6[e]	1.8	80.0	2.0	270.0	Current study
Enzymatic Hydrolysis					
D7	5.5	50.0	48.0	528.0	Current study
D2[f]	nd	nd	nd	38.0	Sigma
D3[g]	nd	nd	nd	510.0	Siddiqui (2014)

nd: Not determined.
[a] pH of dextran solution (20.0 g L^{-1}) was adjusted using H_2SO_4 (0.1N).
[b] Hydrolysis of dextran solution (20.0 g L^{-1}) was carried out at aforementioned temperature.
[c] Time specified for hydrolysis of dextran solution (20.0 g L^{-1}).
[d] Native dextran produced by *L. mesenteroides* KIBGE-IB26.
[e] Hydrolysed fractions of native dextran.
[f] Clinical grade dextran from *L. mesenteroides* NRRL B512F (Sigma, D-4751).
[g] Native dextran from *L. mesenteroides* KIBGE-IB22.

values since the dextran fraction obtained was of 270 kDa. The results showed that acid hydrolysis gave more fractionation range as compared to enzymatic treatment. After the treatment with acid, multiple fractions were collected using different molecular cut off membranes. However, fractional precipitation of dextran can also be performed by organic solvents or ultrafiltration [23]. The chemical analysis of the hydrolyzed fractions of dextran is presented in Table 3.

Initially the samples were analyzed by FTIR spectral analysis in order to examine the type of the linkages and functional groups present in native dextran produced by *L. mesenteroides* KIBGE-IB26 and its hydrolyzed fractions (data not shown). Both native and hydrolyzed products showed similar spectra. Structural analysis of the samples was carried out in the region between 500 cm^{-1} to 4000 cm^{-1}. The signal at 1008.2 cm^{-1}, 1005.8 cm^{-1}, 1003.3 cm^{-1}, 1000.9 cm^{-1} and 993.3 cm^{-1} represents the presence of α-(1 → 6) linkage in D1, D4, D5, D6 and D7, respectively. The signals around this region depict the chain flexibility in the dextran due to the presence of glycosidic bonds [34]. The specific signal peaks representing different functional groups (C—O—C, C—H, C—O,

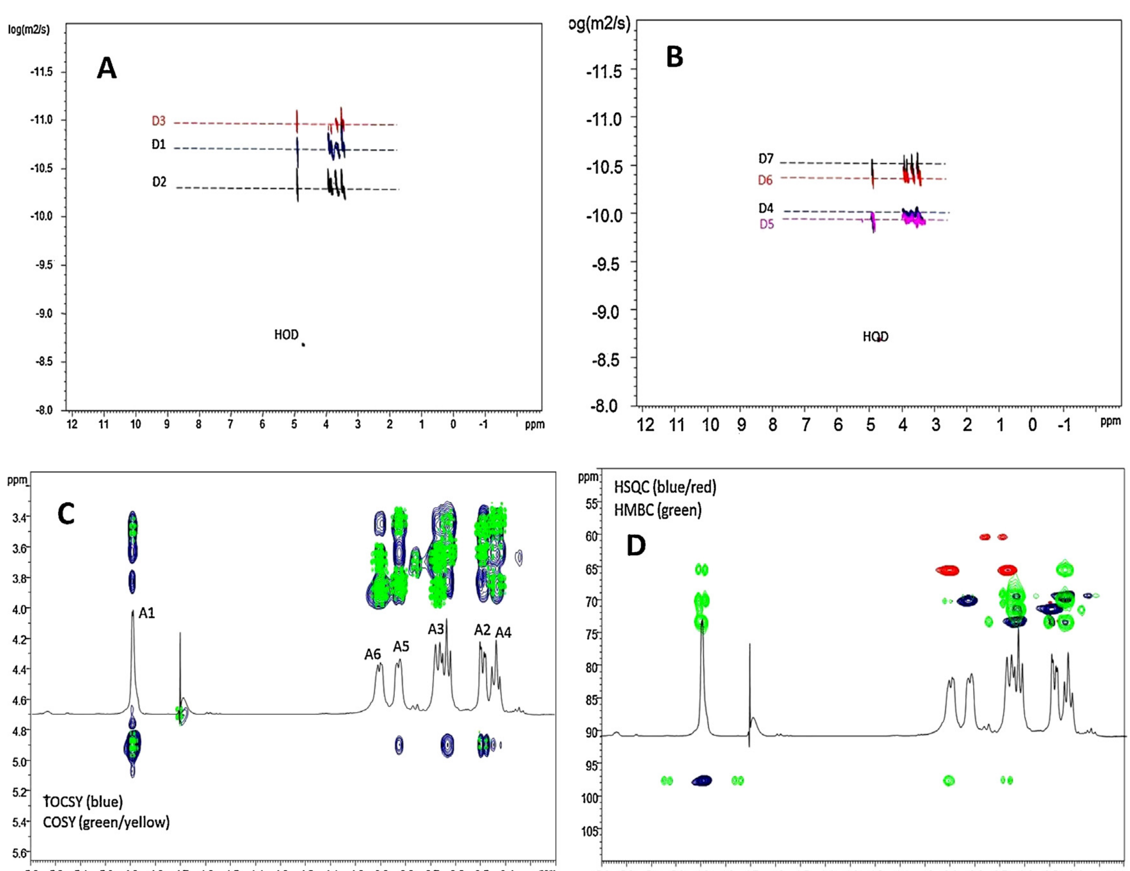

Fig. 1. NMR spectroscopic analysis of native and hydrolyzed low molecular weight fractions.
A and **B:** Molecular weight determination of dextran using diffusion-ordered spectroscopy (DOSY); **C:** Total correlation spectroscopy (TOCSY) & Homonuclear correlation spectroscopy (COSY); **D:** Heteronuclear single-quantum correlation spectroscopy (HSQC) & Heteronuclear multiple-bond correlation spectroscopy (HMBC).

Table 3
Chemical analysis of native and different fractions of dextran.

Sample	Total Sugar[a] (%)	Total Protein[b] (%)	Reducing sugar[c] (%)	Optical Rotation[d] $[\alpha]_D^{20}$
D1	78 ± 3.9	3.3 ± 0.16	0.0005±	+3.50°
Acid Hydrolysis				
D4	80 ± 4.0	2.5 ± 0.12	0.003 ± 0.00001	nd
D5	82 ± 4.1	1.6 ± 0.08	0.0005 ± 0.00002	nd
D6	80 ± 4.0	2.5 ± 0.12	0.0005 ± 0.00002	nd
Enzymatic Hydrolysis				
D7	81 ± 4.0	2.5 ± 0.12	0.0005 ± 0.00002	nd
D3	79 ± 3.9	3.8 ± 0.19	0.36 ± 0.01	nd

nd: Not determined.
[a] Total sugar (g) in 100.0 g of dextran.
[b] Total protein (g) in 100.0 g of dextran.
[c] Reducing sugar(g) in 100.0 g of dextran.
[d] Reading taken on ATAGO Polax 2L.

OH stretching) relating to dextran were noticed and were almost similar in all samples. The absorbance peak at 1147.6 cm^{-1} was due to the covalent vibration of C—O—C bond and glycosidic bridge [34]. The hydroxyl stretching vibration of dextran was recorded in the region of 3286.8 cm^{-1} while; in all other hydrolyzed products the hydroxyl stretching was in same region. C—H stretching vibration presented the band around 2993.5 cm^{-1} regions and a peak at 1650 cm^{-1} was observed due to the existence of carboxyl group in dextran [35,36].

The average molecular weight of native and different dextran fractions (D1–D7) were estimated using Diffusion Ordered NMR Spectroscopy (DOSY) experiments. It is well established that the diffusion coefficient of a certain molecule in a solution, under specific conditions (solvent and/or temperature), is closely related to its molecular size, as shown in the Stokes – Einstein equation:

$$D = kT/6\pi h r_s (m^2\,s^{-1})$$

Where, k is the Boltzman constant; T is the temperature; η is the viscosity of the liquid and r_s represents hydrodynamic radius of the molecule.

It is worth to note that above equation, from which the analysis of DOSY is derived, holds for spherical molecules; however, previous studies demonstrated that diffusion data can provide an estimation of the molecular weight also of uncharged linear and branched oligo- and polysaccharides [8,37]. Therefore, in order to evaluate the molecular weight of the dextran fractions, 2D DOSY experiments were performed on dilute aqueous solutions. The extraction of diffusion coefficient was carried out by the processing software, which evaluates the attenuation of the amplitudes of NMR signals. The resulting processed spectra are reported in Fig. 1A and 1B. A 2D DOSY spectrum exhibits the NMR chemical shifts in one dimension (X-axis) and self-diffusion coefficient in the other one (y-axis). The solvent (D2O) was used as an internal standard; in all the three spectra, the log D value of the HDO signal, at 4.715 ppm, remains constant at −8.686 log(m2/s), indicating that there were no changes in the viscosity of the samples. The signals of each dextran fraction spread along x-axis (chemical shifts dimension) and are characterized by different average log D values. The interpolation of diffusion data is derived from the DOSY spectra according to Viel and coworkers [37].

By using the calibration curve resulting from their study on pullulans as reference, the following molecular weights for each dextran fraction were yielded: D1: 892 kDa; D2 38 kDa; D3 510 kDa; D4 77 kDa; D5 55 kDa; D6 270 kDa and D7 528 kDa. The structural analysis of dextran fractions was then performed by a combination of 1D and 2D NMR spectroscopy (^{1}H NMR, COSY, TOCSY, HSQC and HMBC). All these experiments revealed the presence, in all dextran samples, of two anomeric signals, one resonating at 4.9 ppm (spin system A), characteristic for a typical α-(1 → 6) glycosidic linkage, and one proton signal resonating at 5.27 ppm which is due to the presence of a branch chain typical for α-(1 → 3) linkage (Fig. 1C and 1D). These 2D NMR findings are similar to the data reported earlier [32,38]. The scalar correlation analysis in 2D NMR spectra also permitted the identification of polysaccharide nature. Downfield shift of carbon resonance indicated the glycosylated position at O-6 of the spin system-A. Presences of long range correlations in HMBC spectrum confirm the homopolymer nature of dextran with α-(1 → 6) linked glucose residues.

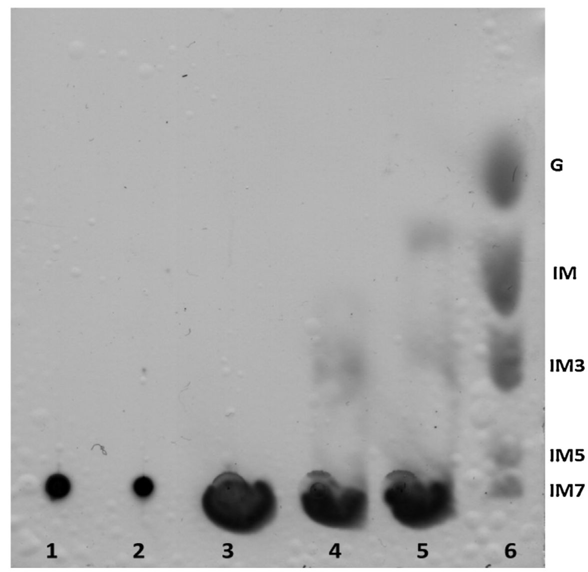

Fig. 2. Thin layer chromatography of hydrolyzed dextran after dextranase treatment.
Lane 1: After 24 h; **2**: After 48 h; **3**: After 72 h; **4**: After 96 h; **5**: After 120 h; **6**: Standard (G: Glucose; IM: Isomaltose; IM3: Isomaltotriose; IM5: Isomaltopentose; IM7: Isomaltoheptose).

When the formation of end products after the treatment with dextranase was studied, it was revealed that the hydrolysis of high molecular weight dextran is highly time dependent. No product was formed up to 72 h of treatment signifying that the dextranase requires more time in order to hydrolyze branched chains typical for α-(1 → 3) linkage present in dextran. However, after 96 h, only isomaltotriose was noticed as compared to 120 h where; isomaltose was also formed along with isomaltotriose (Fig. 2). Similar findings are reported by Vettori, Franchetti and Contiero [32], but the time required was around 14 h. Several other reports also state the formation of high concentrations of isomalto-oligosaccharides

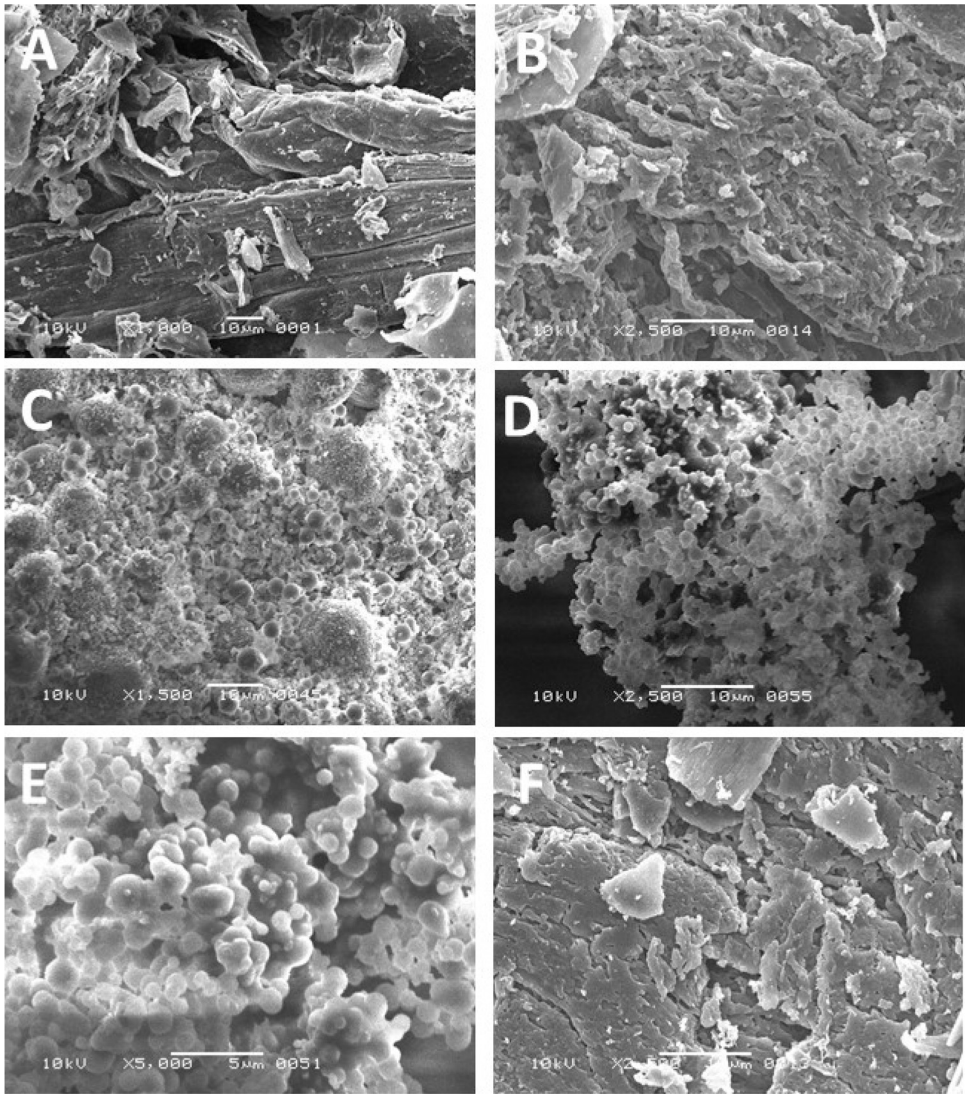

Fig. 3. Scanning electron microscopy (SEM) of native and hydrolyzed low molecular weight dextran.
A and **B:** Native dextran produced by *L. mesenteroides* KIBGE-IB26; **C**: Dextran sample D4; **D**: Dextran sample D5; **E**: Dextran sample D6; **F**: Dextran sample D7.

(isomaltotriose, tetraose, pentose and hexose) as compared to the formation of glucose [39–42].

Scanning electron micrograph of the native and hydrolyzed dextran fractions displayed the similar kind of surface topography of dextran samples as reported earlier [16]. All the dextran samples have porous web like structure in which monomers are closely linked to each other (Fig. 3). Owing to the small pore size and compact structural matrix, it can embrace substantial amount of water molecules. Therefore, the native dextran can be used as texturing agent in food products, while, low molecular weight dextran fraction can be utilized as a carrier vector for the development of a drug delivery system or for other pharmaceutical products.

4. Conclusion

The fermentation parameters for the production of native dextran produced by an indigenous strain of *L. mesenteroides* KIBGE-IB26 were optimized. Maximum yield of the native high molecular weight dextran was obtained using 100.0 g L^{-1} of sucrose after 18 h of incubation at 25 °C by batch fermentation with pH-7.0. Low molecular weight fractions attained after the treatment with dilute acid and enzyme were structurally characterized. Acidic treatment was more effective as compared to the enzymatic one as the fractions obtained showed dispersed molecular weight range instead of clinical grade fractions. Current study revealed that the process of acid hydrolysis is more dependent on pH, as compared to exposure time and temperature. Therefore, low molecular weight fractions could be prepared by acid hydrolysis in shorter time period as compare to enzymatic hydrolysis which usually takes extended time for native dextran to be hydrolyzed and ultimately affects the process cost. However, acid hydrolysis require more purification steps.

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